Sam Shoni

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Summary

Final-year B.Tech student in Robotics and Automation specializing in **ROS 2**, **SLAM** and autonomous navigation. Builds end-to-end robotics systems from Gazebo/RViz simulation to embedded control on Arduino/ESP32 using **Python**, **C/C++** and **OpenCV**. Maintains 50+ public repositories and a portfolio at samshoni.github.io with work on autonomous docking, SLAM, sensor fusion, robotic arms, self-balancing robots and vision-driven automation.

Education

Saintgits College of Engineering (Autonomous), Kerala, India *Expected 2026*
B.Tech in Robotics and Automation Engineering (Final Year)
Current CGPA: **7.4 / 10**

Skills

Core Technologies: ROS 2, SLAM, Nav2, Gazebo, RViz, OpenCV, EKF, PID control
Programming: Python, C/C++, Arduino C, basic shell scripting
Robotics and Middleware: ROS 2 Humble, SLAM Toolbox, Navigation (Nav2), URDF/Xacro
Embedded and Hardware: Arduino, ESP32, Raspberry Pi, IMU (MPU6050), motor drivers (L298N), DC and stepper motors, lidar, encoders, ultrasonic sensors
Tools and Platforms: Linux/Ubuntu, Git and GitHub, VS Code, terminal/CLI, Gazebo + ROS 2 simulation workflows

Projects

6-DOF Robotic Arm with Custom GUI *Arduino, Processing, Servos*
github.com/samshoni/Automated-6DOF-Robotic-Arm-with-GUI

- Designed and built a fully functional 6-DOF robotic arm using servo-based actuation.
- Developed a custom GUI with sliders for manual control and automatic joint sequencing.
- Implemented a record-and-playback system that stores joint positions and repeats them smoothly.
- Integrated Arduino with serial communication for real-time command handling and stable movement.

SLAM & Autonomous Navigation (ROS 2 + Turtlebot3) *ROS 2, Nav2, Gazebo*
github.com/samshoni/my_slam_robot

- Built and ran SLAM on Turtlebot3 in Gazebo to generate a real-time map of the environment.
- Saved the map and configured the Nav2 stack for autonomous navigation with obstacle avoidance.
- Successfully executed 2D Nav Goals with smooth motion planning and dynamic path updates.
- Demonstrated practical understanding of the full ROS2 pipeline: LIDAR data, TF frames, costmaps, and navigation parameters.

Self-Balancing Robot *Arduino, MPU6050, PID Control*
Two-wheeled Closed-loop System

- Built a two-wheeled self-balancing robot using BO motors on a custom chassis.
- Used an MPU6050 IMU to measure tilt angle and detect balance deviation via raw processed output.
- Designed and tuned custom PID values to continuously correct the robot's posture.
- Demonstrated closed-loop feedback behaviour and real-time balance correction using an L298 motor driver.

Multi-Sensor Fusion with Extended Kalman Filter *ROS 2, Python, EKF*
github.com/samshoni/Extended-Kalman-Filter-ros2-sensor-fusion

- Developed a custom ROS 2 package to demonstrate sensor fusion by simulating noisy sensor data.
- Wrote custom Python nodes (`fake_imu_publisher`, `fake_odom_publisher`) to generate synthetic IMU and odometry streams.
- Configured the EKF parameters via YAML to fuse these inputs, optimizing state estimation covariances.

- Validated the fusion logic by comparing the stable fused output trajectory against the raw noisy inputs.

Adaptive Driver-Focus System (Under Patent Preparation) *Python, OpenCV, Real-time*
Real-time Driver Monitoring System

- Developed a real-time driver monitoring system using a standard webcam and facial landmark detection.
- Tracked Eye Aspect Ratio (EAR), blink rate, and head orientation to detect drowsiness and distraction.
- Triggered on-screen alerts when unsafe patterns (long eye closure or head tilt) were detected.
- Built a clean Python/OpenCV pipeline focusing on lightweight, real-time performance.

Experience

Intern, I HUB Robotics – Kochi *4-day Internship*
ROS 2 Humble, SLAM, Nav2, URDF

- Completed hands-on training on ROS 2 Humble: workspace setup, nodes, topics and package structuring.
- Worked with SLAM and Nav2 for mapping and navigation in simulated environments.
- Modeled robots with URDF/Xacro and integrated them with Gazebo and RViz.

Trainee, PSG College of Technology – Coimbatore *1-week Industrial Robotics Training*
FANUC Industrial Robots

- Trained on **FANUC industrial robots**: teaching pendant operation, coordinate frames and basic motion programming.
- Executed pick-and-place and safe manipulator operations on real industrial robots.
- Passed the FANUC assessment and received an industrial robotics certificate.

Certifications & Activities

- **NPTEL** – Completed course on *Mechanics of Solids*.
- **FANUC Industrial Robotics Certificate** – PSG College of Technology, after 1-week hands-on training.
- **IEEE Student Member** – Member of IEEE Student Branch; participated in an IEEE hackathon at Sahrdaya College (certificate awarded).
- **Magazine Club** – Active member, Saintgits College of Engineering.
- **Project Exhibition** – Presented a 6-DOF robotic arm at a technical exhibition at GEC Barton Hill.
- **50+ repositories** on GitHub covering robotics, ROS 2, computer vision and embedded systems.